

Course: Cloud and Network Security CNS-C2-2026

Student Name: James Kirika

Student No: CS-CNS12-26028

September 21, 2026

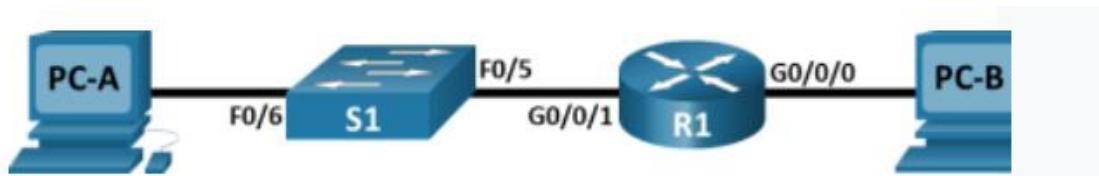
Week 2 Assignment 2:

Build a Switch and Router Network

Table of Contents

Topology.....	3
Addressing table.....	3
Objectives.....	3
Introduction	3
Part 1: Set Up Topology and Initialize Devices	4
Step 1: Cable the network as shown in the topology	4
Step 2: Initialize and reload the router and switch.....	4
Part 2: Configure Devices and Verify Connectivity	5
Step 1: Assign static IP information to the PC interfaces.....	5
Step 2: Configure the router	6
Part 3: Display Device Information	8
Step 1: Display the routing table on the router.	8
a. Use the show ip route	8
b. Use the show ipv6 route	9
Step 2: Display interface information on the router R1.....	10
Use the show ip interface g0/0/1	10
For the IPv6 information, enter the show ipv6 interface.....	11
Step 3: Display a summary list of the interfaces on the router and switch.....	12
Enter the show ip interface brief	12
To see the IPv6 interface information	12
Enter the show ip interface brief command on the switch S1.....	13
Reflection Questions	13
Conclusion.....	13

Topology



Addressing table

Device	Interface	IP Address / Prefix	Default Gateway
R1	G0/0/0	192.168.0.1 /24	N/A
		2001:db8:acad::1/64	
		fe80::1	
	G0/0/1	192.168.1.1 /24	N/A
		200:db8:acad:1::1/64	
		fe80::1	
S1	VLAN 1	192.168.1.2 /24	192.168.1.1
PC-A	NIC	192.168.1.3 /24	192.168.1.1
		2001:db8:acad:1::3/64	fe80::1
PC-B	NIC	192.168.0.3 /24	192.168.0.1
		2001:db8:acad::3/64	fe80::1

Objectives

Part 1: Set Up the Topology and Initialize Devices

Part 2: Configure Devices and Verify Connectivity

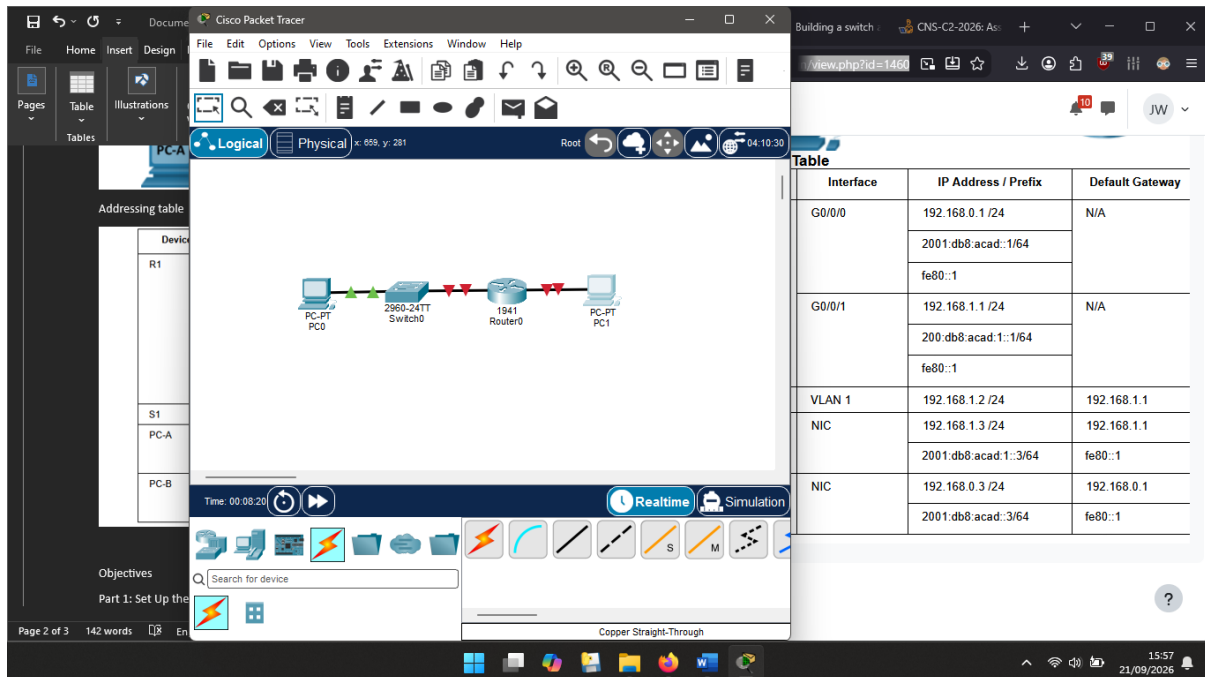
Introduction

This comprehensive lab reviews previously covered IOS commands. The lab required cabling a switch, router, and two PCs according to a topology diagram, then configuring each device to match a provided addressing table. Once configurations is saved, connectivity is verified and IOS commands are used to retrieve device information to answer questions

about the network. Required resources included a Cisco 4221 router, a Cisco 2960 switch, two PCs with terminal emulation software, console cables, and Ethernet cables.

Part 1: Set Up Topology and Initialize Devices

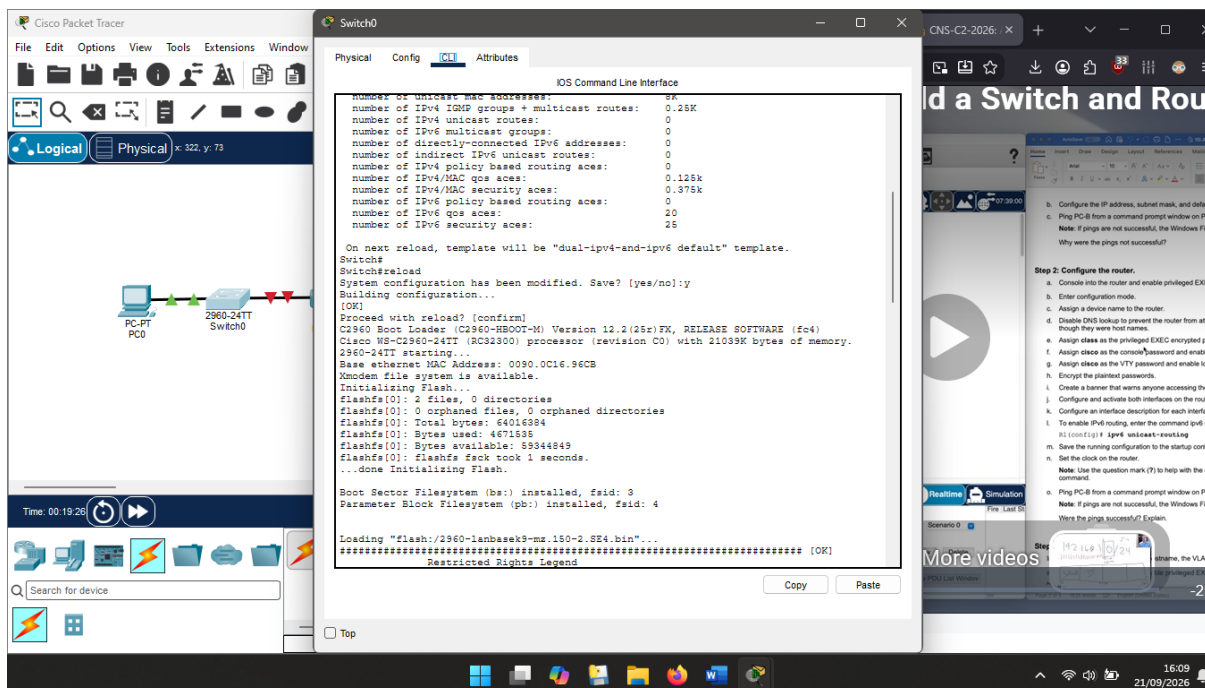
Step 1: Cable the network as shown in the topology.



The image shows the Cisco Packet Tracer interface with a network topology and a configuration table. The topology consists of a 2960-24TT switch connected to a 1941 router, which is then connected to two PC-PT PCs. The configuration table provides the IP addresses and default gateways for each interface.

Interface	IP Address / Prefix	Default Gateway
G0/0/0	192.168.0.1 /24	N/A
	2001.db8.acad:1/64	
	fe80::1	
G0/0/1	192.168.1.1 /24	N/A
	2001.db8.acad:1:1/64	
	fe80::1	
VLAN 1	192.168.1.2 /24	192.168.1.1
NIC	192.168.1.3 /24	192.168.1.1
	2001.db8.acad:1:3/64	fe80::1
NIC	192.168.0.3 /24	192.168.0.1
	2001.db8.acad:3/64	fe80::1

Step 2: Initialize and reload the router and switch.



The image shows the Cisco Packet Tracer interface with the terminal output for the switch and router initialization. The switch is being reloaded, and the router is being initialized. The terminal output shows the boot process, including the loading of the IOS image and the initialization of the flash file system.

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Number of unicast mac addresses: 8K
number of IPv4 IGMP groups + multicast routes: 0.25K
number of IPv4 unicast routes: 0
number of IPv6 multicast groups: 0
number of directly-connected IPv6 addresses: 0
number of indirect IPv6 unicast routes: 0
number of IPv4 policy based routing aces: 0
number of IPv4/MAC qos aces: 0.125k
number of IPv4/MAC security aces: 0.375k
number of IPv6 policy based routing aces: 0
number of IPv6 qos aces: 20
number of IPv6 security aces: 25
On next reload, template will be "dual-ipv4-and-ipv6 default" template.
Switch0
Switch#reload
System configuration has been modified. Save? [yes/no]:y
Building configuration...
[OK]
Proceed with reload? [confirm]
C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25)FX, RELEASE SOFTWARE (fc4)
Cisco WS-C2960-24TT (RC32300) processor (revision C0) with 21039K bytes of memory.
2960-24TT starting...
Base ethernet MAC Address: 0090.0C16.96C8
Xmodem file system is available.
Initializing Flash...
flashfs[0]: 2 files, 0 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 64016584
flashfs[0]: Bytes used: 4671535
flashfs[0]: Bytes available: 59344849
flashfs[0]: flashfs fsck took 1 seconds.
...done Initializing Flash.
Boot Sector Filesystem (bs:) installed, fsid: 3
Parameter Block Filesystem (pb:) installed, fsid: 4
Loading "flash:/2960-lanbasek9-mz.150-2.SF4.bin"...
*****
Restricted Rights Legend
Copy Paste
```

Part 2: Configure Devices and Verify Connectivity

Step 1: Assign static IP information to the PC interfaces.

Configure the IP address, subnet mask, and default gateway settings on PC-A.

The screenshot shows the Cisco Packet Tracer interface with PC-A configuration open. The IP Configuration tab is active, showing static IP settings for both IPv4 and IPv6. The IPv4 settings are: IP Address 193.168.1.3, Subnet Mask 255.255.255.0, Default Gateway 192.168.1.1, and DNS Server 0.0.0.0. The IPv6 settings are: IPv6 Address 2001:db8:acad:1::3 / 64, Link Local Address FE80::260:3EFF:FE60:AC8E, Default Gateway fe80::1, and DNS Server. The 802.1X section is also visible.

On the right, a network diagram shows the following configuration:

Interface	IP Address	Subnet Mask	Default Gateway
G0/0/1	192.168.1.1 / 24	N/A	
VLAN 1	192.168.1.2 / 24	192.168.1.1	
NIC	192.168.1.3 / 24	192.168.1.1	
NIC	192.168.0.3 / 24	192.168.0.1	

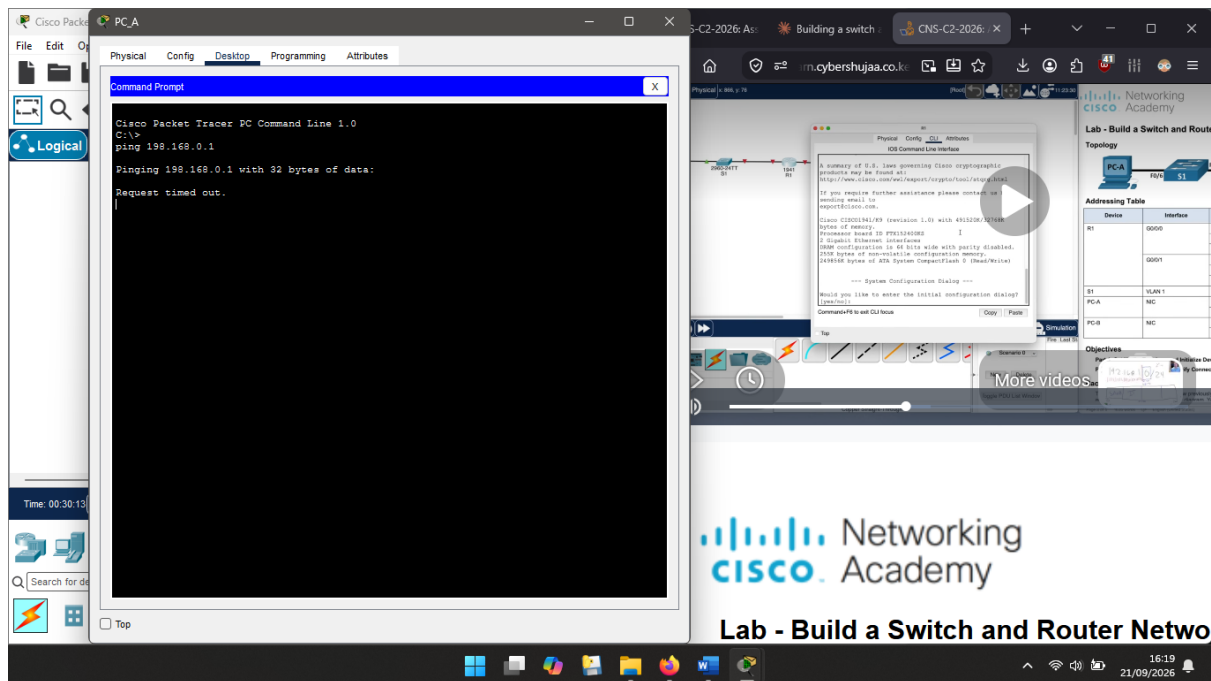
Configure the IP address, subnet mask, and default gateway settings on PC-B.

The screenshot shows the Cisco Packet Tracer interface with PC-B configuration open. The IP Configuration tab is active, showing static IP settings for both IPv4 and IPv6. The IPv4 settings are: IP Address 192.168.0.3, Subnet Mask 255.255.255.0, Default Gateway 192.168.0.1, and DNS Server 0.0.0.0. The IPv6 settings are: IPv6 Address 2001:db8:acad:3::3 / 64, Link Local Address FE80::260:21FF:FE27:506C, Default Gateway fe80::1, and DNS Server.

On the right, a network diagram shows the following configuration:

Interface	IP Address	Subnet Mask	Default Gateway
G0/0/1	192.168.1.1 / 24	N/A	
VLAN 1	192.168.1.2 / 24	192.168.1.1	
NIC	192.168.1.3 / 24	192.168.1.1	
NIC	192.168.0.3 / 24	192.168.0.1	

Ping PC-B from a command prompt window on PC-A.

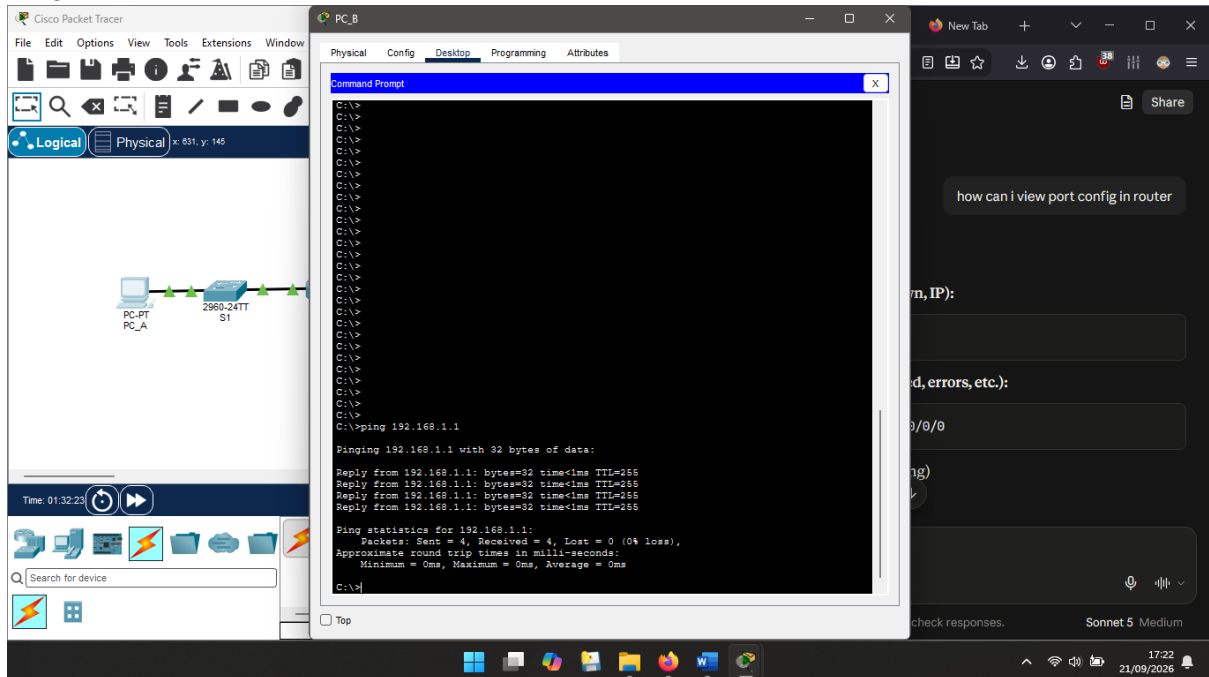


Got a request time out. It failed because the switch and the router are haven't been setup yet.

Step 2: Configure the router

- Console into the router and enable privileged EXEC mode.
- Enter configuration mode.
- Assign a device name to the router.
- Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- Assign class as the privileged EXEC encrypted password.
- Assign cisco as the console password and enable login.
- Assign cisco as the VTY password and enable login.
- Encrypt the plaintext passwords.
- Create a banner that warns anyone accessing the device that unauthorized access is prohibited
- Configure and activate both interfaces on the router.

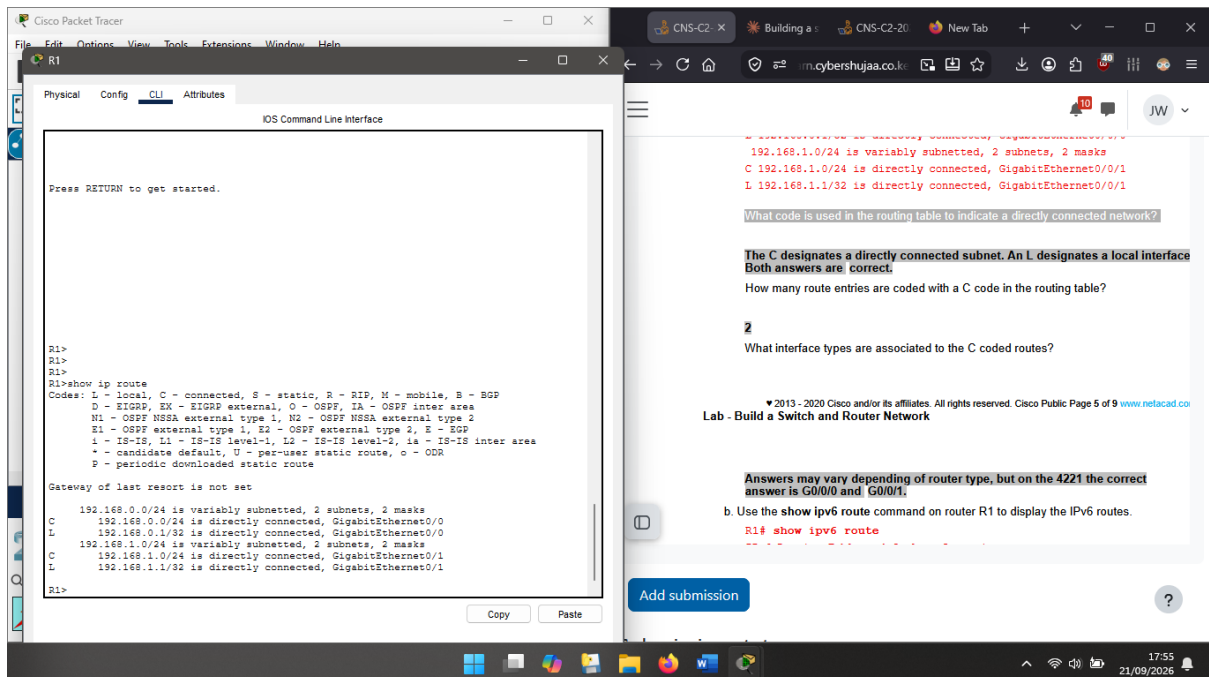
Ping PC_A from B cmd



Part 3: Display Device Information

Step 1: Display the routing table on the router.

a. Use the show ip route



Query

What code is used in the routing table to indicate a directly connected network? The C designates a directly connected subnet. An L designates a local interface.

What interface types are associated to the C coded routes?

Both C (connected) routes are on GigabitEthernet interfaces:

- 192.168.0.0/24 → GigabitEthernet0/0
- 192.168.1.0/24 → GigabitEthernet0/1

b. Use the show ipv6 route

IOS Command Line Interface

```
R1>
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
C       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1

R1>
R1>show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       H - ISIS L1, L2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D -
       EIGRP EX - EIGRP external, ND - ND Default, NDp - ND Prefix,
       DCE - Destination, NDR - Redirect, RL - RPL, O - OSPF Intra, OI
       - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-
       registrations ld - LISP dyn-eid, la - LISP away, le -
       LISP extranet-policy a - Application
C 2001:DB8:ACAD::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L 2001:DB8:ACAD::1/128 [0/0]
    via GigabitEthernet0/0, receive
C 2001:DB8:ACAD::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L 2001:DB8:ACAD::1/128 [0/0]
    via GigabitEthernet0/1, receive
L FE80::8 [0/0]
    via Null0, receive

R1>
```

Answers may vary depending of router type, but on the 4221 the correct answer is G0/0/0 and G0/0/1.

b. Use the **show ipv6 route** command on router R1 to display the IPv6 routes.

R1# show ipv6 route

IPv6 Routing Table - default - 5 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination, NDR - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter

OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1

ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations ld - LISP dyn-eid, la - LISP away, le - LISP extranet-policy a - Application

C 2001:DB8:ACAD::/64 [0/0]

Step 2: Display interface information on the router R1.

Use the show ip interface g0/0/1

The screenshot shows the Cisco Packet Tracer interface on the left and a web browser on the right. The Packet Tracer window displays the CLI of router R1 with the command `show ip interface g0/0/1` executed. The output shows the interface is up and connected, with an IP address of 192.168.1.1/24. The web browser window shows the same output, with the command `show ip interface g0/0/1` highlighted in red. Below the output, there is a section titled "Step 2: Display interface information on the router R1." with a sub-question: "Use the show ip interface g0/0/1 to answer the following questions. R1# show ip interfaces g0/0/1". The output of the command is displayed, showing the interface is up and connected, with an IP address of 192.168.1.1/24.

The screenshot shows the Cisco Packet Tracer interface on the left and a web browser on the right. The Packet Tracer window displays the CLI of router R1 with the command `show interface g0/0/1` executed. The output shows the interface is up and connected, with a description of "Link to PC Network". The web browser window shows the same output, with the command `show interface g0/0/1` highlighted in red. Below the output, there is a section titled "What is the operational status of the G0/0/1 interface?" with a sub-question: "What is the Media Access Control (MAC) address of the G0/0/1 interface?". The output of the command is displayed, showing the interface is up and connected, with a description of "Link to PC Network".

query

What is the operational status of the G0/0/1 interface?

GigabitEthernet0/0/1 is up, line protocol is up (connected)

Both the physical layer (interface) and data link layer (line protocol) are up — the interface is fully operational.

What is the Media Access Control (MAC) address of the G0/1 interface?

0010.11d0.7c02

How is the Internet address displayed in this command?

Internet address is 192.168.1.1/24

For the IPv6 information, enter the show ipv6 interface

The screenshot shows a Cisco Packet Tracer simulation environment. On the left, a router (R1) is shown with its configuration window open. The configuration window displays the following output:

```
Output queue 0/0/0 (size/max):
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 4 bits/sec, 0 packets/sec
10 packets input, 1120 bytes, 0 no buffer
Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 watchdog, 1017 multicast, 0 pause input

R1>show ipv6 int?
^
Unrecognized command
R1>show ipv6 interface
^
Invalid input detected at '^' marker.

R1>show ipv6 in?
inspect interface
R1>show ipv6 in
R1>show ipv6 interface
GigabitEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::1
No Virtual link-local address(es):
Global unicast address(es):
2001:DB8:ACAD::1, subnet is 2001:DB8:ACAD::/64
Joined group address(es):
FF02::1
FF02::2
FF02::1:FE00:1
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachables are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
ND advertised reachable time is 0 (unspecified)
ND advertised neighbor interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
GigabitEthernet0/1 is up, line protocol is up
--More--
```

On the right, a web browser window is open, displaying the output of the `show ipv6 interface` command. The output is as follows:

```
x will be replaced with a hexadecimal number.
How is the Internet address displayed in this command?

Internet address is 192.168.1.1/24.

b For the IPv6 information, enter the show ipv6 interface
interface command. R1# show ipv6 interface
g0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::1
No Virtual link-local address(es):
Description: Connected to F0/5 on S1
Global unicast address(es):
2001:DB8:ACAD::1, subnet is 2001:DB8:ACAD::/64
Joined group address(es):
FF02::1
FF02::2
FF02::1:FE00:1
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachables are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
--More--
```

Step 3: Display a summary list of the interfaces on the router and switch.

Enter the show ip interface brief

The screenshot shows a Cisco Packet Tracer simulation environment. On the left, the 'CLI' window for router R1 is open, displaying the following commands and output:

```
R1>show ipv6 int?  
% Unrecognized command  
R1>show ipv6 interface  
% Invalid input detected at '^' marker.  
R1>show ipv6 in?  
% Invalid input detected at '^' marker.  
R1>show ipv6 in  
% Invalid input detected at '^' marker.  
R1>show ipv6 interface  
GigabitEthernet0/0 is up, line protocol is up  
IPv6 is enabled, link-local address is FE80::1  
No Virtual link-local address(es):  
Global unicast address(es):  
2001:DB8:ACAD::1, subnet is 2001:DB8:ACAD::/64  
Joined group address(es):  
FF02::1  
FF02::2  
FF02::1:FE80::1  
MTU is 1500 bytes  
ICMP error messages limited to one every 100 milliseconds  
ICMP redirects are enabled  
ICMP unreachable are sent  
ND DAD is enabled, number of DAD attempts: 1  
ND reachable time is 30000 milliseconds  
ND advertised reachable time is 0 (unspecified)  
ND advertised retransmit interval is 0 (unspecified)  
ND router advertisements are sent every 200 seconds  
ND router advertisements live for 1800 seconds  
ND advertised default router preference is Medium  
Hosts use stateless autoconfig for addresses.  
GigabitEthernet0/1 is up, line protocol is up  
R1>show ip interface brief  
Interface IP-Address OK? Method Status Protocol  
GigabitEthernet0/0 192.168.0.1 YES manual up up  
GigabitEthernet0/1 192.168.1.1 YES manual up up  
Vlan1 unassigned YES unset administratively down down  
R1>
```

On the right, a web browser window displays a Netacad lab page titled 'Lab - Build a Switch and Router Network'. The page includes the following text:

Step 3: Display a summary list of the interfaces on the router and switch.

There are several commands that can be used to verify an interface configuration. One of the most useful of these is the **show ip interface brief** command. The command output displays a summary list of the interfaces on the device and provides immediate feedback to the status of each interface.

a. Enter the **show ip interface brief** command on the router R1.

```
R1# show ip interface brief  
Interface IP-Address OK? Method Status Protocol  
GigabitEthernet0/0/0 192.168.0.1 YES manual up up  
GigabitEthernet0/0/1 192.168.1.1 YES manual up up  
Serial0/1/0 unassigned YES unset up up  
Serial0/1/1 unassigned YES unset up up
```

b. To see the IPv6 interface information, enter the **show ipv6 interface brief** command on R1. R1# **show ipv6 interface brief**

```
GigabitEthernet0/0/0 [up/up]  
FE80::1  
2001:DB8:ACAD::1  
GigabitEthernet0/0/1 [up/up]  
FE80::1  
2001:DB8:ACAD::1  
Serial0/1/0 [up/up]
```

To see the IPv6 interface information

The screenshot shows a Cisco Packet Tracer simulation environment. On the left, the 'CLI' window for router R1 is open, displaying the following commands and output:

```
R1>show ip interface brief  
Interface IP-Address OK? Method Status Protocol  
GigabitEthernet0/0 192.168.0.1 YES manual up up  
GigabitEthernet0/1 192.168.1.1 YES manual up up  
Vlan1 unassigned YES unset administratively down down  
R1>  
R1>  
R1>  
R1>  
R1>  
R1>show ipv6 interface brief  
GigabitEthernet0/0 [up/up]  
FE80::1  
2001:DB8:ACAD::1  
GigabitEthernet0/1 [up/up]  
FE80::1  
2001:DB8:ACAD::1  
Vlan1 [administratively down/down]  
unassigned  
R1>
```

On the right, a web browser window displays a Netacad lab page titled 'Lab - Build a Switch and Router Network'. The page includes the following text:

a. Enter the **show ip interface brief** command on the router R1.

```
R1# show ip interface brief  
Interface IP-Address OK? Method Status Protocol  
GigabitEthernet0/0/0 192.168.0.1 YES manual up up  
GigabitEthernet0/0/1 192.168.1.1 YES manual up up  
Serial0/1/0 unassigned YES unset up up  
Serial0/1/1 unassigned YES unset up up
```

b. To see the IPv6 interface information, enter the **show ipv6 interface brief** command on R1. R1# **show ipv6 interface brief**

```
GigabitEthernet0/0/0 [up/up]  
FE80::1  
2001:DB8:ACAD::1  
GigabitEthernet0/0/1 [up/up]  
FE80::1  
2001:DB8:ACAD::1  
Serial0/1/0 [up/up]  
unassigned  
Serial0/1/1 [up/up]  
unassigned  
GigabitEthernet0 [down/down]  
Unassigned
```

c. Enter the **show ip interface brief** command on the switch S1.

```
S1# show ip interface brief
```

Enter the show ip interface brief command on the switch S1.

IOS Command Line Interface

```
Press RETURN to get started.
```

```
S1>
S1>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	down	down
FastEthernet0/2	unassigned	YES	manual	down	down
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	up	up
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down

Copy Paste

Enter the show ip interface brief command on the switch S1

```
S1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol	Vlan1
192.168.1.2	YES	NVRAM	up	FastEthernet0/1	unassigned	YES
down	down	FastEthernet0/2	unassigned	YES	unset	down
down	down	FastEthernet0/3	unassigned	YES	unset	down
down	down	FastEthernet0/4	unassigned	YES	unset	down
down	down	FastEthernet0/5	unassigned	YES	unset	down
down	down	FastEthernet0/6	unassigned	YES	unset	down
down	down	FastEthernet0/7	unassigned	YES	unset	down
down	down	FastEthernet0/8	unassigned	YES	unset	down
down	down	FastEthernet0/9	unassigned	YES	unset	down
down	down	FastEthernet0/10	unassigned	YES	unset	down
down	down	FastEthernet0/11	unassigned	YES	unset	down
down	down	FastEthernet0/12	unassigned	YES	unset	down
down	down	FastEthernet0/13	unassigned	YES	unset	down

Reflection Questions

1. If the G0/0/1 interface showed that it was administratively down, what interface configuration command would you use to turn the interface up?
no shutdown
2. What would happen if you had incorrectly configured interface G0/0/1 on the router with an IP address of 192.168.1.2?
PC-A would not be able to ping PC_B, because PC_A is configured to use the IP address of 192.168.1.1 for the default-gateway router, but this address is not assigned to any device on the LAN. Any packets that need to be sent to the default-gateway for routing will never reach their destination.

Conclusion

This lab reinforced core cisco skills; cabling, addressing, and verifying a switch-and-router network. Commands like show ip route, show ip interface brief, and show interfaces showed how to check routing, Layer 1/2 status, IP addressing, and MAC addresses.